

CENG381 Stochastic Processes

Biased Random Walk — Practice 1

Q1 (30 points). A particle starts at position $k = 2$ on the integer path $\{0, 1, 2, 3, 4\}$. State 0 is a 'Pit of Disaster' (absorbing) and state 4 is 'Home' (absorbing). At each step the particle moves right with probability $p = 2/3$ and left with probability $q = 1/3$.

- Let $R_n = P(\text{reach Home} \mid \text{start at } n)$. Write the recurrence relation for R_n ($1 \leq n \leq 3$) and state boundary conditions R_0 and R_4 .
 - The characteristic equation of the homogeneous part is $pr^2 - r + q = 0$. Factor it as $(pr - q)(r - 1) = 0$ to find both roots. Compute q/p and write the general solution $R_n = \alpha_1 (q/p)^n + \alpha_2$.
 - Apply $R_0 = 0$ and $R_4 = 1$ to find α_1 and α_2 . Compute R_2 . Compare with the unbiased result $R_2 = 2/4 = 1/2$ and explain the direction of the change.
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Q2 (35 points). Using the same biased walk ($p = 2/3$, $q = 1/3$) on $\{0, 1, 2, 3, 4\}$, let $X_n = E[\text{absorption time} \mid \text{start at } n]$. Both barriers 0 and 4 are absorbing, so $X_0 = X_4 = 0$.

- Write the recurrence for X_n ($1 \leq n \leq 3$). Note this is a linear non-homogeneous recurrence.
 - The homogeneous solution is $\alpha_1 (1/2)^n + \alpha_2$. Find a particular solution of the form $cn + d$ by substituting into the recurrence. Show that $c = -1/(p-q) = -3$ and $d = 0$.
 - Write the general solution $X_n = \alpha_1 (1/2)^n + \alpha_2 - 3n$. Apply $X_0 = 0$ and $X_4 = 0$ to find α_1 and α_2 . Compute X_2 .
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Q3 (35 points). A stencil starts at position $k = 3$ on the path $\{0, 1, 2, 3, 4, 5, 6\}$ ($w = 6$). State 0 = 'Pit of Disaster' and state 6 = 'Cliff of Doom', both absorbing. The walk is biased: $p = 2/3$ (right) and $q = 1/3$ (left).

- Find $R_n = P(\text{reach Cliff of Doom} \mid \text{start } n)$ using the formula:

$$R_n = \frac{[(q/p)^n - 1]}{[(q/p)^w - 1]}$$

Compute R_3 . Does the particle more likely end up in the Pit or at the Cliff?

- For the 'Cliff of Doom' boundary conditions ($R_0 = 1, R_w = 0$), the formula becomes:

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$$R_n = \frac{(q/p)^n - (q/p)^w}{1 - (q/p)^w}$$

Compute R_3 under these conditions. Verify that the two probabilities from (a) and (b) sum to 1.

c) The expected absorption time follows $X_n = \alpha_1 \cdot (1/2)^n + \alpha_2 - 3n$ with $X_0 = X_6 = 0$. Find X_3 . Compare with the unbiased expected time $X_n^{\text{(unbiased)}} = n(w - n)$, which gives $X_3^{\text{(unbiased)}} = 9$.